# Exercise sheet 2 – Explanation

Linear Optimization (WS 26/27)

**Erklärung**

Dieses Blatt behandelt drei Grundideen: Gleichungssysteme mit mehr Variablen als Gleichungen, die Standardform aus der Vorlesung und graphisches Lösen linearer Programme. Bei der Standardform im Skript wird ein Maximierungsproblem mit Gleichungen $Ax = b$ und Nichtnegativität $x \geq 0$ verwendet.

## Classroom task 2.1

*What happens to the solution* $(x_1, x_2)$ *of the linear system* $Ax = b$ *with* $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$ *and* $b = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ *if we add a column to the matrix on the left hand side, so that* $\widetilde{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}$*? How can we represent the solution to the new system?*

**Erklärung**

Zuerst lösen wir das alte System. Danach bekommt das neue System eine zusätzliche Variable $x_3$. Weil dann zwei Gleichungen und drei Variablen vorhanden sind, wird eine Variable als freier Parameter gewählt.

The original system is

$$\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

It is equivalent to

$$\begin{aligned} x_1 + x_2 &= 2 && \text{row 1 of } Ax = b \\ 3x_2 &= 3 && \text{row 2 of } Ax = b \end{aligned}$$

Solving the second equation gives

$$\begin{aligned} 3x_2 &= 3 && \text{second equation} \\ x_2 &= 1 && \mid /3 \end{aligned}$$

Substitute this value into the first equation:

$$\begin{aligned} x_1 + x_2 &= 2 && \text{first equation} \\ x_1 + 1 &= 2 && \text{insert } x_2 = 1 \\ x_1 &= 1 && \mid -1 \end{aligned}$$

**Erklärung**

Das alte System hat genau eine Lösung, weil die beiden Gleichungen zwei Variablen festlegen.

Thus the original solution is

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The new system is

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

It is equivalent to

$$\begin{aligned} x_1 + x_2 &= 2 && \text{row 1 of } \widetilde{A}x = b \\ 3x_2 + x_3 &= 3 && \text{row 2 of } \widetilde{A}x = b \end{aligned}$$

Let

$$x_3 = t \qquad \text{choose the new variable as a parameter}$$

Then the second equation gives

$$\begin{aligned} 3x_2 + x_3 &= 3 && \text{second equation} \\ 3x_2 + t &= 3 && \text{insert } x_3 = t \\ 3x_2 &= 3 - t && | - t \\ x_2 &= 1 - \frac{t}{3} && | /3 \end{aligned}$$

Now the first equation gives

$$\begin{aligned} x_1 + x_2 &= 2 && \text{first equation} \\ x_1 + 1 - \frac{t}{3} &= 2 && \text{insert } x_2 = 1 - \frac{t}{3} \\ x_1 - \frac{t}{3} &= 1 && | - 1 \\ x_1 &= 1 + \frac{t}{3} && | + \frac{t}{3} \end{aligned}$$

**Erklärung**

Der Parameter $t$ kann jede reelle Zahl sein. Daher ist die neue Lösungsmenge eine Gerade im dreidimensionalen Raum.

Check:

$$\begin{aligned} x_1 + x_2 &= 1 + \frac{t}{3} + 1 - \frac{t}{3} && \text{insert } x_1 \text{ and } x_2 \\ &= 2 && \frac{t}{3} - \frac{t}{3} = 0 \\ 3x_2 + x_3 &= 3\left(1 - \frac{t}{3}\right) + t && \text{insert } x_2 \text{ and } x_3 \\ &= 3 - t + t && \text{distribute } 3 \\ &= 3 && -t + t = 0 \end{aligned}$$

$$\boxed{\left\{ \begin{pmatrix} 1 + \frac{t}{3} \\ 1 - \frac{t}{3} \\ t \end{pmatrix} \middle| \, t \in \mathbb{R} \right\}}$$

## Exercise 2.1

*Consider the linear optimization problem* $\max\{2x_1 + x_2 \mid 2x_1 - x_2 \leq 12,\ 0 \leq x_1 \leq 8,\ x_2 \geq 0\}$. *Transform the problem into standard form and explicitly define* $c$, $b$ *and* $A$ *as in the lecture notes. Is the problem feasible? Is the problem bounded?*

**Erklärung**

Die Vorlesungs-Standardform lautet: maximize $c^T x$ subject to $Ax = b$ and $x \geq 0$. Deshalb werden die Ungleichungen mit Schlupfvariablen in Gleichungen umgeschrieben.

The lecture standard form is

$$\begin{aligned} &\text{maximize} && c^T x \\ &\text{subject to} && Ax = b, \\ & && x \geq 0. \end{aligned}$$

The constraint $0 \leq x_1 \leq 8$ gives nonnegativity of $x_1$ and the upper bound

$$x_1 \leq 8 \qquad \text{upper bound from } 0 \leq x_1 \leq 8$$

Introduce slack variables $s_1, s_2 \geq 0$:

$$\begin{aligned} 2x_1 - x_2 + s_1 &= 12 && \text{add slack variable } s_1 \text{ to the first inequality} \\ x_1 + s_2 &= 8 && \text{add slack variable } s_2 \text{ to the upper bound} \end{aligned}$$

Define

$$y = (x_1, x_2, s_1, s_2)^T.$$

Then

$$A = \begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 12 \\ 8 \end{pmatrix}, \qquad c = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

**Erklärung**

Die Nullen in $c$ stehen dort, weil die Schlupfvariablen nur Nebenbedingungen ausgleichen und nicht in der ursprünglichen Zielfunktion vorkommen.

The standard form is

$$\boxed{\begin{aligned} &\text{maximize} && 2x_1 + x_2 \\ &\text{subject to} && 2x_1 - x_2 + s_1 = 12, \\ & && x_1 + s_2 = 8, \\ & && x_1, x_2, s_1, s_2 \geq 0. \end{aligned}}$$

Feasibility check: Choose

$$y = (0, 0, 12, 8)^T.$$

$$\begin{aligned} 2 \cdot 0 - 0 + 12 + 0 &= 12 && \text{first row of } Ay \\ 0 + 8 &= 8 && \text{second row of } Ay \end{aligned}$$

Thus $Ay = b$. Also

$$y \geq 0 \qquad \text{all entries } 0, 0, 12, 8 \text{ are nonnegative}$$

Therefore the problem is feasible.

Boundedness check: For every $M \geq 0$, choose

$$y(M) = (0, M, 12 + M, 8)^T.$$

Then

$$\begin{aligned} 2 \cdot 0 - M + 12 + M &= 12 && \text{first row of } Ay(M) \\ 0 + 8 &= 8 && \text{second row of } Ay(M) \end{aligned}$$

Thus $Ay(M) = b$. The objective value is

$$\begin{aligned} c^T y(M) &= 2 \cdot 0 + M + 0 \cdot (12 + M) + 0 \cdot 8 && \text{insert } y(M) \\ &= M && \text{terms with factor 0 vanish} \end{aligned}$$

**Erklärung**

Da $M$ beliebig groß gewählt werden kann, werden auch die Zielfunktionswerte beliebig groß. Das ist bei Maximierung genau Unbeschränktheit.

The problem is feasible and unbounded.

## Exercise 2.2

*Determine the linear program that was solved graphically in Figure 1. Standard form is not required. The figure shows a feasible region to the right of* $x_1 = 3$*, above the line through* $(0, 4)$ *and* $(6, 0)$*, and above the line through* $(1, 0)$ *and* $(7, 3)$*. The red objective lines have slope* $-2$ *and touch the feasible region first at* $(3, 2)$*.*

**Erklärung**

Aus einer gezeichneten Geraden lesen wir zwei Punkte ab. Daraus wird ihre Gleichung bestimmt. Bei einer Zielfunktionsgeraden $c_1 x_1 + c_2 x_2 = \alpha$ ist die Steigung $-c_1/c_2$. Die rote Steigung $-2$ passt zu $2x_1 + x_2 = \alpha$.

The vertical boundary gives

$$x_1 \geq 3 \qquad \text{feasible region is to the right of the line } x_1 = 3$$

The blue boundary passes through $(0, 4)$ and $(6, 0)$. Its slope is

$$\begin{aligned} m &= \frac{0 - 4}{6 - 0} && \text{slope formula} \\ &= -\frac{4}{6} && \text{compute numerator and denominator} \\ &= -\frac{2}{3} && \text{divide numerator and denominator by 2} \end{aligned}$$

Thus the blue line has equation

$$\begin{aligned} x_2 &= 4 - \frac{2}{3} x_1 && \text{intercept 4 and slope } -\frac{2}{3} \\ 3x_2 &= 12 - 2x_1 && \mid \cdot 3 \\ 2x_1 + 3x_2 &= 12 && \mid + 2x_1 \end{aligned}$$

The feasible region is above that line, hence

$$2x_1 + 3x_2 \geq 12 \qquad \text{choose the side above the blue line}$$

The rising boundary passes through $(1, 0)$ and $(7, 3)$. Its slope is

$$\begin{aligned} m &= \frac{3 - 0}{7 - 1} && \text{slope formula} \\ &= \frac{3}{6} && \text{compute numerator and denominator} \\ &= \frac{1}{2} && \text{divide numerator and denominator by 3} \end{aligned}$$

Using the point $(1, 0)$ gives

$$\begin{aligned} x_2 - 0 &= \frac{1}{2}(x_1 - 1) && \text{point-slope form} \\ x_2 &= \frac{1}{2}x_1 - \frac{1}{2} && \text{distribute } \frac{1}{2} \\ 2x_2 &= x_1 - 1 && \mid \cdot 2 \\ -x_1 + 2x_2 &= -1 && \mid -x_1 \end{aligned}$$

The feasible region is above that line, hence

$$-x_1 + 2x_2 \geq -1 \qquad \text{choose the side above the rising line}$$

The red objective lines have slope $-2$, so one matching objective is

$$z = 2x_1 + x_2 \qquad \text{line } 2x_1 + x_2 = \alpha \text{ has slope } -2$$

Because the shaded region is unbounded to the upper right, the graph solves a minimization problem. Therefore one equivalent linear program represented by the figure is

$$\begin{array}{ll} \text{minimize} & 2x_1 + x_2 \\ \text{subject to} & x_1 \geq 3, \\ & 2x_1 + 3x_2 \geq 12, \\ & -x_1 + 2x_2 \geq -1. \end{array}$$

Check the marked point:

$$\begin{aligned} 2 \cdot 3 + 3 \cdot 2 &= 6 + 6 && \text{insert } (3, 2) \text{ into } 2x_1 + 3x_2 \\ &= 12 && 6 + 6 = 12 \\ 3 &\geq 3 && \text{vertical constraint} \\ -3 + 2 \cdot 2 &= -3 + 4 && \text{insert } (3, 2) \text{ into } -x_1 + 2x_2 \\ &= 1 && -3 + 4 = 1 \\ 1 &\geq -1 && \text{compare} \end{aligned}$$

The objective value at $(3, 2)$ is

$$\begin{aligned} z &= 2 \cdot 3 + 2 && \text{insert } (3, 2) \\ &= 6 + 2 && 2 \cdot 3 = 6 \\ &= 8 && 6 + 2 = 8 \end{aligned}$$

The graphical optimum is $x^* = (3, 2)$ with value 8.

## Exercise 2.3

*Solve the following linear program graphically: maximize* $3x_1 + 5x_2$ *subject to* $x_1 + 2x_2 \leq 12$, $3x_1 + x_2 \leq 16$, $0 \leq x_2 \leq 4$, *and* $x_1 \geq 0$.

**Erklärung**

Beim graphischen Lösen reichen die Eckpunkte der zulässigen Menge: Bei einem linearen Programm nimmt eine lineare Zielfunktion ihr Optimum, falls es existiert, an einem Eckpunkt an. Wir bestimmen die Eckpunkte und vergleichen dort die Zielfunktionswerte.

The boundary lines are

| | |
|---|---|
| $x_1 + 2x_2 = 12$ | first inequality as equality |
| $3x_1 + x_2 = 16$ | second inequality as equality |
| $x_2 = 0$ | lower bound for $x_2$ |
| $x_2 = 4$ | upper bound for $x_2$ |
| $x_1 = 0$ | lower bound for $x_1$ |

Find the intersection of the two slanted boundary lines:

| | |
|---|---|
| $x_1 + 2x_2 = 12$ | first boundary |
| $3x_1 + x_2 = 16$ | second boundary |

From the first boundary,

| | |
|---|---|
| $x_1 + 2x_2 = 12$ | first boundary |
| $x_1 = 12 - 2x_2$ | $\mid -2x_2$ |

Insert this into the second boundary:

| | |
|---|---|
| $3x_1 + x_2 = 16$ | second boundary |
| $3(12 - 2x_2) + x_2 = 16$ | insert $x_1 = 12 - 2x_2$ |
| $36 - 6x_2 + x_2 = 16$ | distribute 3 |
| $36 - 5x_2 = 16$ | $-6x_2 + x_2 = -5x_2$ |
| $-5x_2 = -20$ | $\mid -36$ |
| $x_2 = 4$ | $\mid /(-5)$ |

Then

| | |
|---|---|
| $x_1 = 12 - 2x_2$ | formula from first boundary |
| $x_1 = 12 - 2 \cdot 4$ | insert $x_2 = 4$ |
| $x_1 = 12 - 8$ | $2 \cdot 4 = 8$ |
| $x_1 = 4$ | $12 - 8 = 4$ |

So one vertex is $(4, 4)$.
The vertex on $x_2 = 0$ and $3x_1 + x_2 = 16$ is found by

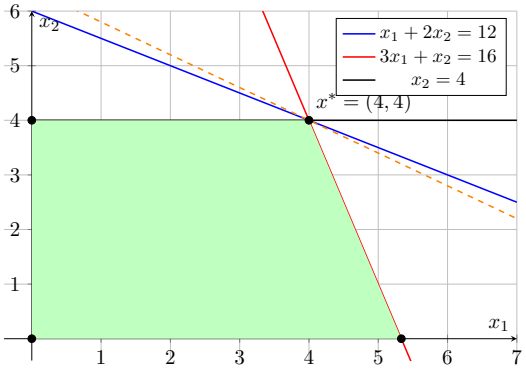
| | |
|---|---|
| $3x_1 + x_2 = 16$ | second boundary |
| $3x_1 + 0 = 16$ | insert $x_2 = 0$ |
| $3x_1 = 16$ | $3x_1 + 0 = 3x_1$ |
| $x_1 = \frac{16}{3}$ | $\mid /3$ |

Thus this vertex is $\left(\frac{16}{3}, 0\right)$.
The remaining axis vertices are

| | |
|---|---|
| $(0, 0)$ is feasible | both lower bounds active |
| $(0, 4)$ is feasible | $x_1 = 0$ and upper bound $x_2 = 4$ active |

**Erklärung**

Der Punkt $(0, 6)$ aus $x_1 + 2x_2 = 12$ ist nicht zulässig, weil $x_2 \leq 4$ gilt. Daher liegt der obere linke Eckpunkt bei $(0, 4)$.




Evaluate the objective function $z = 3x_1 + 5x_2$ at the feasible vertices:

$$
\begin{aligned}
z(0,0) &= 3 \cdot 0 + 5 \cdot 0 && \text{insert } (0,0)\\
&= 0 && \text{compute}\\
z\left(\frac{16}{3}, 0\right) &= 3 \cdot \frac{16}{3} + 5 \cdot 0 && \text{insert } \left(\frac{16}{3}, 0\right)\\
&= 16 + 0 && 3 \cdot \frac{16}{3} = 16\\
&= 16 && 16 + 0 = 16\\
z(4,4) &= 3 \cdot 4 + 5 \cdot 4 && \text{insert } (4,4)\\
&= 12 + 20 && 3 \cdot 4 = 12 \text{ and } 5 \cdot 4 = 20\\
&= 32 && 12 + 20 = 32\\
z(0,4) &= 3 \cdot 0 + 5 \cdot 4 && \text{insert } (0,4)\\
&= 0 + 20 && 3 \cdot 0 = 0 \text{ and } 5 \cdot 4 = 20\\
&= 20 && 0 + 20 = 20
\end{aligned}
$$

**Erklärung**

Der größte Wert in der Liste ist 32. Die gestrichelte orange Zielfunktionsgerade im Bild ist $3x_1 + 5x_2 = 32$ und berührt die zulässige Menge im Punkt $(4, 4)$.

Check feasibility of the optimal point:

$$
\begin{aligned}
4 + 2 \cdot 4 &= 4 + 8 && \text{insert } (4,4) \text{ into } x_1 + 2x_2\\
&= 12 && 4 + 8 = 12\\
12 &\leq 12 && \text{first constraint is satisfied}\\
3 \cdot 4 + 4 &= 12 + 4 && \text{insert } (4,4) \text{ into } 3x_1 + x_2\\
&= 16 && 12 + 4 = 16\\
16 &\leq 16 && \text{second constraint is satisfied}\\
0 &\leq 4 && \text{lower bound for } x_2\\
4 &\leq 4 && \text{upper bound for } x_2\\
4 &\geq 0 && \text{lower bound for } x_1
\end{aligned}
$$

$$\boxed{x^* = (4, 4), \qquad z^* = 32}$$